Ecuación de Clairaut

$$y = x\frac{dy}{dx} + Q\left(\frac{dy}{dx}\right)$$

$$\frac{dy}{dx} = p$$

$$y = xp + \Phi p$$

$$\frac{dy}{dx} = p + x\frac{dp}{dx} + Q'(p)\frac{dp}{dx}$$

$$\frac{dp}{dx}[x + Q'(p)] = 0$$

$$\frac{dp}{dx} = 0 \therefore p = c$$

$$y = cx + Q(c)$$

$$y = x\frac{dy}{dx} - \ln\frac{dy}{dx}$$

$$p = \frac{dy}{dx}$$

$$y = xp - \ln p$$

$$\frac{dy}{dx} = x\frac{dp}{dx} + p - \frac{dp}{p\,dx}$$

$$p = x\frac{dp}{dx} + p - \frac{dp}{p\,dx}$$

$$0 = \frac{dp}{dx}\left(x - \frac{1}{p}\right)$$

$$\frac{dp}{dx} = 0$$

$$p = c$$

$$y = cx - \ln c$$

Calcular la envolvente

$$x - \frac{1}{p} = 0 \qquad p = \frac{1}{x}$$

$$y = \frac{x}{x} - \ln\left(\frac{1}{x}\right)$$

$$y = 1 + \ln x$$

1) $\left(\frac{dy}{dx}\right)^2 - x\frac{dy}{dx} + y = 0$

$$x\frac{dy}{dx} - \left(\frac{dy}{dx}\right)^2$$

$$y = x\frac{dy}{dx} - \left(\frac{dy}{dx}\right)$$

$$p = \frac{dy}{dx}$$

$$y = xp - p^2$$

$$\frac{dy}{dx} = x\frac{dp}{dx} + p - 2p\frac{dp}{dx}$$

$$\cancel{p} = x\frac{dp}{dx} + \cancel{p} - 2p\frac{dp}{dx}$$

$$0 = \frac{x\,dp}{dx} - 2p\frac{dp}{dx}$$

$$0 = (x - 2p)\frac{dp}{dx}$$

$$\frac{dp}{dx} = 0$$

$$p = c$$

$$y = xc - c^2$$

$$(x - 2p) = 0$$

$$x = 2p$$

$$p = \frac{x}{2}$$

$$y = x\frac{x}{2} - \left(\frac{x}{2}\right)^2$$

$$y = \frac{x^2}{2} - \frac{x^2}{4}$$

$$y = \frac{x^2}{4}$$